

# Tharusha Sihan Fonseka

Biomedical Engineering Undergraduate, University of Moratuwa

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## EDUCATION

### University of Moratuwa

B.Sc. Engineering (Hons.) in Biomedical Engineering; **CGPA: 3.94/4.00** [🔗](#)

Moratuwa, Sri Lanka

Jun 2021 - Present

#### Dean's List: Semesters 1-7

- Robotics, Autonomous Systems, Electronic Control Systems, Advanced Electronic Control Systems.
- Electronic Instrumentation, Embedded Systems, Medical Electronics and Instrumentation, Electronic Circuit Design.
- Digital Signal Processing, Bio Signal Processing, Medical Imaging, Data Structures and Algorithms.

#### Minor in Mathematics

- Numerical Methods, Linear Algebra, Number Theory and Cryptography, Advanced Differential Equations, Linear Models and Multivariate Statistics, Calculus.

### Ananda College

GCE Advanced Level - Physical Science Stream; Z-Score 3.1996; **Island Rank -1<sup>st</sup>**

Colombo 10, Sri Lanka

Jan 2017 - Aug 2019

- Combined Mathematics, Physics, Chemistry

#### Online courses

|                                                                  |          |
|------------------------------------------------------------------|----------|
| Computer Vision (Kaggle)                                         | May 2025 |
| Intro to Deep Learning (Kaggle)                                  | May 2025 |
| ROS2 for Beginners (Udemy)                                       | Apr 2025 |
| Robotics: Aerial Robotics (University of Pennsylvania, Coursera) | Oct 2023 |
| Machine Learning Specialization (DeepLearning.AI, Coursera)      | Apr 2023 |

## EXPERIENCE

### Visiting Instructor - University of Moratuwa

Feb 2025 - May 2025

- Conducting electronic laboratory practicals of EN1094 - Laboratory practice module.

### Research Affiliate - University of Sydney

Dec 2023 - May 2024

- Simulated the stretching of epidermal sensors with different body motions to analyze the stress and strain.
- Designed a wearable device to capture and transmit real-time resistance data from an epidermal stretch sensor.
- Setup the hardware for simultaneous data capturing from two radar devices in different frequency bands to create a synthetic band using machine learning.

## PROJECTS

### Humanoid Head For Robot Receptionist Capable of Natural Human Interaction Jun 2024 - May 2025

- Project team lead.
- The robot head is capable of bio mimicking facial expressions, active eye gazing, human re-identification, and natural conversation abilities.
- I worked on hardware design, implementation, and ROS2-based controlling.

### Phoenix Cube Satellite Project

Ongoing

- Stage 1, launching a high-altitude balloon into the stratosphere.
- I am leading a team of twenty undergraduates.

### Re-implementing research paper on Bio Signal Processing [🔗](#)

2024

- Re-implemented the paper "Classification of ECG heartbeats using nonlinear decomposition methods and support vector machine".

### Maze Solving Micromouse Robot(Robofest 2023, 2024)🏆

2023, 2024

- Fully Autonomous maze solving robot, which is tasked to reach the destination point in the shortest time possible.
- I contributed to the sensor fusion, control system, and implementation of the maze-solving algorithm.

### Scorpion - Battle Robot (EXMO 2023) 🏆

Jul 2023

- Designed a battle robot to demonstrate in EXMO 2023.
- I was involved in designing and implementing the mechanical and structural design.

### Smart Switch for Automating Conventional Home Appliances ☯

Jun 2023

- Designed a smart switch to automate conventional home appliances, which can be easily integrated into an existing domestic installation.
- Can be controlled remotely through internet or manually through capacitive touch buttons.
- An individual project.

### Analog Implementation of a Five Band Audio Equalizer ☯

Feb 2023

- Five band audio equalizer using analog 4<sup>th</sup> order Sallen Key filters and a differential amplifier.

### IV Fluid Flow Regulator ☯

Aug 2022

- An automatic IV fluid flow regulator to aid in the treatment of dengue patients.
- Flow rate and the period can be selected by the user.
- Fluid is pumped using a peristaltic rotary pump.

## AWARDS AND SCHOLARSHIPS

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### Winners | Electronic Design Competition

Sep 2023

IEEE Sri Lanka section

### 1<sup>st</sup> Runners-up | SPARK Challenge 2023

Jun 2023

Innovation Competition for Sustainable Development - Electronic Club of University of Moratuwa

### Dialog Merit Scholarship Award

2023

For outstanding performance in GCE A/L Examination

### S.A. Wijethilake Memorial Challenge Cup | Ananda College

2020

Awarded for the Best Academic Performance of the College at the G.C.E. A/L Examination 2019

### “Anada Pradeepa” Award | Ananda College

2020

Awarded to the College subject toppers based on the G.C.E. A/L Examination 2019

### Silver medalist | Sri Lankan Physics Olympiad

2019

Awarded by Institute of Physics Sri Lanka

### Silver medalist | Sri Lankan Astronomy and Astrophysics Olympiad

2018

Awarded by Institute of Physics Sri Lanka

## LEADERSHIP AND VOLUNTEERING

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### Project Manager - SEDS-Mora - University of Moratuwa

Apr 2024 - Apr 2025

### Old Anandian Engineers' Guild - Students' Chapter

Secretary

Mar 2024 - Apr 2025

Vice President

Mar 2023 - Mar 2024

Deputy Director- School Affairs

Jan 2022 - Mar 2023

### Event Team Lead - Brainstorm 2023 - IEEE EMBS, University of Moratuwa

Aug 2022 - May 2023

### Field Representative - Department of ENTC - University of Moratuwa

May 2022 - Feb 2023

### Event Co-chair - TourEye - Roteract Club - University of Moratuwa

Sept 2021

### Chief Organizer - Anandian Astronomical Association - Ananda College

Nov 2017 - Nov 2018

## SKILLS SUMMARY

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**Languages:** English, Sinhala (Native Proficiency)

**Programming Languages:** Python, Matlab, C++

**Frameworks:** ROS2

**Software:** PCB designing (Altium), CAD designing (Solidworks), Circuit simulation (Proteus, LtSpice)

**Soft Skills:** Leadership, Project Management, Team working

## REFERENCES

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### Dr. Peshala Jayasekara

B.Sc.Eng.Hons(Moratuwa), M.Eng(UTokyo),

Ph.D(UTokyo), MIEEE

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### Dr. Anusha Withana

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